

Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

Ch 7 · Discrete Fourier Transform and FFT

Numerical Problems & Solutions

82 of the 99 questions are two calculations: an N -point DFT by DIT or DIF (43 questions, 340 marks) and a circular convolution (39 questions, 253 marks). The other 17 are bookwork, answered in the theory chapter. Every computational paper in the 45 is below.

The DIT recipe

1. Pad $x[n]$ with zeros up to N , a power of 2. Eleven papers print six or seven values for an 8-point transform; **that is the zero padding, not a misprint.**
2. Reorder the input into bit-reversed order. For $N = 8$ that is $x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]$.
3. Stage 1: $N/2$ butterflies on *adjacent* pairs, twiddle $W_N^0 = 1$, so it is pure add and subtract.
4. Stage 2: butterflies spanning 2, twiddles W_N^0 and W_N^2 .
5. Stage 3: butterflies spanning 4, twiddles $W_N^0 \dots W_N^3$.
6. Read the output in natural order: it is $X[0]$ to $X[N-1]$.

The DIF recipe, which is the same thing backwards

1. Pad to N as before, but **leave the input in natural order.**
2. Stage 1: butterflies spanning $N/2$, twiddle applied to the *difference*, twiddles $W_N^0 \dots W_N^3$.
3. Stage 2: span $N/4$, twiddles W_N^0 and W_N^2 .
4. Stage 3: span 1, twiddle W_N^0 , pure add and subtract.
5. **Unscramble the output:** what comes out in position i is $X[\text{bitrev}(i)]$.

The twiddle factors, which is all the arithmetic there is

W_8^k	Exact	Decimal
W_8^0	1	1
W_8^1	$\frac{1}{\sqrt{2}}(1-j)$	0.7071 – j0.7071
W_8^2	$-j$	$-j$
W_8^3	$-\frac{1}{\sqrt{2}}(1+j)$	–0.7071 – j0.7071

And W_4^k is 1, $-j$, -1 , $+j$. Beyond $k = 3$ use $W_8^{k+4} = -W_8^k$; **there are only four numbers to know.**

Checks that cost nothing

1. $X[0] = \sum x[n]$. One addition, and it catches a wrong first stage.
2. $X[N-k] = X^*[k]$ for real $x[n]$. Compute $X[1]$ to $X[N/2]$ and **mirror the rest**, which halves the work and is not a shortcut the examiner can object to.
3. $X[N/2] = \sum (-1)^n x[n]$, the alternating sum.
4. Parseval: $\sum |x[n]|^2 = \frac{1}{N} \sum |X[k]|^2$.
5. For a convolution, $\sum y[n] = (\sum x_1)(\sum x_2)$.

1 The 8-point DFT by DIT and by DIF

TOP 43/45 80 Bh, 81 Ba, 82 Ba, 80 Ba, 70 Ch, 78 Bh, 79 Bh, 81 Bh, 79 Ba, 76 Ch, 76 Ash, 82 Bh, 75 Ch, 75 Ash, 74 Ch, 74 Ash, 70 Bh, 81 Ch, 76 Bh, 70 Ma, 73 Ch, 73 Shr, 72 Ch, 72 Ka, 71 Ch, 70 Asa, 71 Shr, 71 Bh, 69 Ch, 80 Ch, 73 Bh, 72 Ma, 78 Ch, 77 Ch, 75 Bh, 74 Bh, 73 Ma, 74 Ma, 66 Ma, 68 Bh, 69 Bh, 72 Ash · **340 marks, the largest single question in the subject**

QUESTION AS PRINTED 2080 Baishakh · 2070 Chaitra · [2+6]

Find the 8-point DFT of the sequence $x[n] = \{1, 1, 0, 0, 1, 1, 2\}$ using the decimation in frequency FFT algorithm. 2080 Baishakh also asks for the same sequence by the decimation in time algorithm.

This is the one sequence a paper asks BOTH ways, so it is worked both ways here. The answers are identical, which is the point.

Step 0: pad, and say so

- Seven values are printed for an 8-point transform, so $x[7] = 0$: $x[n] = \{1, 1, 0, 0, 1, 1, 2, 0\}$.
- Check target: $X[0] = \sum x[n] = 1 + 1 + 0 + 0 + 1 + 1 + 2 + 0 = 6$.

Route A: decimation in time

- Bit-reversed input order 0, 4, 2, 6, 1, 5, 3, 7 gives $\{x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]\} = \{1, 1, 0, 2, 1, 1, 0, 0\}$.

	0	1	2	3	4	5	6	7
input (bit-rev)	1	1	0	2	1	1	0	0
after stage 1	2	0	2	-2	2	0	0	0
after stage 2	4	$j2$	0	$-j2$	2	0	2	0
after stage 3	6	$j2$	$-j2$	$-j2$	2	$j2$	$j2$	$-j2$

- **Stage 1** is pure add/subtract ($W^0 = 1$) on adjacent pairs: $1 \pm 1 = 2, 0$; $0 \pm 2 = 2, -2$; $1 \pm 1 = 2, 0$; $0 \pm 0 = 0, 0$.
- **Stage 2** spans 2, with twiddles $W_8^0 = 1$ and $W_8^2 = -j$. The $-j$ acting on -2 is what produces the $\pm j2$.
- **Stage 3** spans 4, twiddles $W_8^0, W_8^1, W_8^2, W_8^3$. The last row is already $X[k]$ in **natural order**.

Route B: decimation in frequency, same sequence

	0	1	2	3	4	5	6	7
input (natural)	1	1	0	0	1	1	2	0
after stage 1	2	2	2	0	0	0	$j2$	0
after stage 2	4	2	0	$-j2$	$j2$	0	$-j2$	0
after stage 3	6	2	$-j2$	$j2$	$j2$	$j2$	$-j2$	$-j2$
unscrambled	6	$j2$	$-j2$	$-j2$	2	$j2$	$j2$	$-j2$

- **Stage 1** pairs $x[n]$ with $x[n+4]$: sums 2, 2, 2, 0 on top; the differences 0, 0, -2, 0 are then multiplied by $W_8^0, W_8^1, W_8^2, W_8^3$, and $(-2)(W_8^2) = (-2)(-j) = j2$ lands in position 6.
- **The last row is in bit-reversed order.** Position i holds $X[\text{bitrev}(i)]$, so reading positions 0, 4, 2, 6, 1, 5, 3, 7 gives the natural order shown in the final row.

$$X(k) = \{6, j2, -j2, -j2, 2, j2, j2, -j2\}$$

The checks, and a pleasant surprise

- $X[0] = 6$. ✓ $X[4] = \sum (-1)^n x[n] = 1 - 1 + 0 - 0 + 1 - 1 + 2 - 0 = 2$. ✓
- $X[7] = X^*[1]$: $-j2 = (j2)^*$. ✓ And $X[6] = X^*[2]$, $X[5] = X^*[3]$. ✓
- $|X(k)| = \{6, 2, 2, 2, 2, 2, 2, 2\}$: every non-zero frequency has the same magnitude 2, and only the phase differs. Worth writing down, because it looks like an error and is not.
- Parseval: $\sum |x[n]|^2 = 1 + 1 + 0 + 0 + 1 + 1 + 4 + 0 = 8$, and $\frac{1}{8}(36 + 7 \times 4) = \frac{64}{8} = 8$. ✓

Every other FFT paper, question as printed

Same recipe, different sequences, so the answer column is the result rather than a repeat of the three stages. **DIT and DIF give the same $X(k)$** , and each paper's own wording says which it wants. Sequences shorter than N are zero padded; **the padding is never stated by the paper** and is the first mark.

Papers	Question as printed	$X(k)$
80 Bh	Compute 8-point DIF-FFT of sequence $x(n) = \{2, 1, 2, 1, 1, 2, 1, 2\}$. 8	$\{12, 1+j0.4142, 0, 1+j2.414, 0, 1-j2.414, 0, 1-j0.4142\}$
78 Bh	Find the 8 - point DFT of $x[n] = u[n] - u[n-4]$ using FFT DIT algorithm. 7 a formula, not a list: it samples to $\{1, 1, 1, 1, 0, 0, 0, 0\}$	$\{4, 1-j2.414, 0, 1-j0.4142, 0, 1+j0.4142, 0, 1+j2.414\}$
81 Bh	How does FFT reduce computational complexity compared to direct calculation of DFT? Compute 8-point DFT of $x[n] = \{1, -2, -4, 3, 0, 1\}$ using DIT-FFT algorithm. 2+7	$\{-1, -3.243+j4, 5+j4, 5.243-j4, -5, 5.243+j4, 5-j4, -3.243-j4\}$
79 Ba	Why we need DFT? Find 8-point DFT of sequence $x[n] = \{1, 2, 4, 3, 5, -1, 3\}$ using Decimation in Frequency Fast Fourier Transform (DIFFFT) algorithm. 2+6	$\{17, -4-j5.243, -1+j2, -4-j3.243, 9, -4+j3.243, -1-j2, -4+j5.243\}$
76 Ch	Find 8-point DFT of sequence $x[n] = \{1, 2, 3, 3, 5, 0, 4, 6\}$ using Decimation in frequency Fast Fourier Transform (DIFFFT) algorithm. 7	$\{24, -0.4645+j1.707, -1+j7, -7.536-j0.2929, 2, -7.536+j0.2929, -1-j7, -0.4645-j1.707\}$
76 Ash	Why we need DFT? Find 8-point DFT of sequence $x[n] = \{1, 2, 3, 3, 5, 1, 4, 2\}$ using Decimation in frequency Fast Fourier Transform (DIFFFT) algorithm. 2+8	$\{21, -4-j0.4142, -1+j2, -4-j2.414, 5, -4+j2.414, -1-j2, -4+j0.4142\}$
82 Bh	Find 8-point DFT using DIF-FFT of the sequence $x[n] = \{1, \frac{1}{2}, -1, -\frac{1}{2}, 2, -\frac{3}{2}\}$. 8 printed as fractions; same numbers as $\{1, 0.5, -1, -0.5, 2, -1.5\}$	$\{0.5, 0.7678-j0.06066, 4+j0.5, -2.768-j2.061, 3.5, -2.768+j2.061, 4-j0.5, 0.7678+j0.06066\}$
75 Ash	(its FFT marks are on the circular convolution in §2; the sequence $\{1, 2, 3, 4, 5, 6, 7, 8\}$ below is the standard worked one)	$\{36, -4+j9.657, -4+j4, -4+j1.657, -4, -4-j1.657, -4-j4, -4-j9.657\}$
74 Ch	Why do we need DFT? Draw the butterfly structure to compute the DFT of the following signal using Radix-2 DIFFFT algorithm, and compute $X(2)$ and $X(1)$ only $x[n] = \{1.5, -1, 1.8, 0.6, 3, 1.7\}$ 3+7	$\{7.6, -3.833-j0.3151, 2.7-j0.1, 0.8335+j3.285, 5, 0.8335-j3.285, 2.7+j0.1, -3.833+j0.3151\}$
74 Ash, 70 Bh	Compute the 8-point DFT of the sequence $x[n] = \{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\}$ using DIF-FFT. 7 70 Bh prints the same four values as 0.5 and says "radix-2 DIF" 8	$\{2, 0.5-j1.207, 0, 0.5-j0.2071, 0, 0.5+j0.2071, 0, 0.5+j1.207\}$
81 Ch	Why we need FFT? Compute the 8-point DFT of sequence $x(n) = \{-0.5, -0.5, 0.5, 0.5, 0, 0, 0, 0\}$ using Decimation in Frequency Fast Fourier Transform (DIFFFT) algorithm. 2+6 the paper closes the brace with a round bracket	$\{0, -1.207-j0.5, -1+j1, 0.2071+j0.5, 0, 0.2071-j0.5, -1-j1, -1.207+j0.5\}$
73 Ch	Why we need FFT? Find 8-point DFT of sequence $x[n] = \{1, -1, 3, 2, 1, 1, 3, -2\}$ using Decimation in frequency Fast Fourier Transform (DIFFFT) algorithm. 2+6	$\{8, -4.243-j1.414, -4, 4.243-j1.414, 8, 4.243+j1.414, -4, -4.243+j1.414\}$
73 Shr	Why do we need DFT? Find 8-point DFT of sequence $x[n] = \{1, -1, 2, 2, 1, 1, 2, 2\}$ using Fast Fourier Transform algorithm. 2+6 names neither DIT nor DIF , so either is accepted; say which you used	$\{10, -1.414+j1.414, -2+j4, 1.414+j1.414, 2, 1.414-j1.414, -2-j4, -1.414-j1.414\}$
72 Ch	Find the FFT of the signal $x[n] = \{1, 1, 2, 4, 3, 1, 2, 1\}$ using DIT-FFT algorithm. 8	$\{15, -4.121-j2.121, j3, 0.1213-j2.121, 1, 0.1213+j2.121, -j3, -4.121+j2.121\}$
72 Ka	Find 8-point DFT of sequence $x[n] = \{1, 1, 0, 1, 0, 1, 2\}$ using Decimation in Time Fast Fourier Transform (DITFFT) algorithm. 7	$\{6, 0.2929+j1.293, -1-j1, 1.707-j2.707, 0, 1.707+j2.707, -1+j1, 0.2929-j1.293\}$
71 Shr, 71 Bh	Find DFT for $\{1, 1, 2, 0, 1, 2, 0, 1\}$ using FFT DIT butterfly algorithm and plot the spectrum . 6+2 71 Bh prints "using FFT" and the same sequence 8	$\{8, -j0.5858, -j2, j3.414, 0, -j3.414, j2, j0.5858\}$
69 Ch	Why do we need FFT? Find 8-point DFT of sequence $x[n] = \{1, 1, 2, 2, 1, 1, 2, 1\}$ using Decimation in frequency FFT (DIFFT) algorithm. 2+7	$\{11, -0.7071-j0.7071, -2+j1, 0.7071-j0.7071, 1, 0.7071+j0.7071, -2-j1, -0.7071+j0.7071\}$
80 Ch, 73 Bh	Why we need a DFT? Find 8-point DFT of sequence $x[n] = \{1, 0, 2, 0, -1, 1, 1\}$ using Decimation in Time Fast Fourier Transform (DITFFT) algorithm. 1+7 1+6	$\{4, 1.293-j0.2929, -3-j1, 2.707+j1.707, 2, 2.707-j1.707, -3+j1, 1.293+j0.2929\}$
72 Ma	Why we need FFT? Find 8-point DFT of sequence $x[n] = \{1, 0, 0, 0, -1, 1, 1\}$ using Decimation in Frequency Fast Fourier Transform (DIFFFT) algorithm. 1+6	$\{2, 1.293+j1.707, -1-j1, 2.707-j0.2929, 0, 2.707+j0.2929, -1+j1, 1.293-j1.707\}$
74 Bh	Discuss the computational efficiency of FFT algorithm. Find DFT of a sequence given a $x[n] = \{1, 1, 0, 0, 0\}$ using DIT FFT algorithm. 2+5	$\{2, 1.707-j0.7071, 1-j1, 0.2929-j0.7071, 0, 0.2929+j0.7071, 1+j1, 1.707+j0.7071\}$
73 Ma	Discuss the computational efficiency of FFT algorithm. Find DFT of a sequence given as $x[n] = \{1, -1, 2\}$ using DIT FFT algorithm. 2+5	$\{2, 0.2929-j1.293, -1+j1, 1.707+j2.707, 4, 1.707-j2.707, -1-j1, 0.2929+j1.293\}$
74 Ma	Find 8 point DFT of $\{1, 1, 0, -1, 2\}$ using DFTFFT algorithm. 7 "DFTFFT" is a typo for DITFFT	$\{3, 0.4142, 3-j2, -2.414, 3, -2.414, 3+j2, 0.4142\}$

- **Five of these do not hand you a list of eight numbers.** 78 Bh prints a step difference $u[n] - u[n-4]$; 82 Bh prints fractions; 74 Ash prints halves and 70 Bh the same values as decimals. **Sample or convert first, then pad**, and write the padded sequence down before touching a butterfly.

- **74 Ch wants only $X(1)$ and $X(2)$** , not all eight, and says so. The two marked values are the answer; computing the other six is unpaid work. The butterfly *structure* still has to be drawn, which is where its 3 marks are.
- **71 Shr and 71 Bh also want the spectrum plotted.** Plot $|X(k)|$ against k : here $\{8, 0.5858, 2, 3.414, 0, 3.414, 2, 0.5858\}$, symmetric about $k = 4$, which is the conjugate-symmetry check drawn rather than written.
- **73 Shr names no algorithm at all**, only "Fast Fourier Transform". Either is correct; name the one you used in the first line.

1.1 The four-point papers

HOT 8/45 81 Ba **2+6**, 82 Ba **1+2+6**, 75 Ch **2+8**, 76 Bh **2+5**, 70 Ma **7**, 66 Ma **6**, 68 Bh **8**, 69 Bh **5** · **two stages, not three, and the twiddles are only 1 and $-j$**

Paper	Alg	$x[n]$	$X(k)$	Note
81 Ba	DIT	$\{2, 2, 4\}$	$\{8, -2-j2, 4, -2+j2\}$	padded to 4
82 Ba	DIF	$\{1, 4, -1\}$	$\{4, 2-j4, -4, 2+j4\}$	padded to 4
75 Ch	DIT	$\{1, -2, 3, 2\}$	$\{4, -2+j4, 4, -2-j4\}$	asks $X(3)$ and $X(5)$
76 Bh	DIT	$\{0.5, 0.5, 0.5, 0.5\}$	$\{2, 0, 0, 0\}$	constant $\rightarrow$ impulse
70 Ma	DIT	$\{2, 0, 1, 2\}$	$\{5, 1+j2, 1, 1-j2\}$	
66 Ma	DIT	$\{1, 3, 4, 5\}$	$\{13, -3+j2, -3, -3-j2\}$	"show the computation"
68 Bh	DIF	$\{1, -2, 2, 1\}$	$\{2, -1+j3, 4, -1-j3\}$	
69 Bh	DIT	$\{3.6, 5.5, 3.3, 6.3\}$	$\{18.7, 0.3+j0.8, -4.9, 0.3-j0.8\}$	

- **66 Ma worked, since it says "show the computation"**. Bit-reversed input $\{x[0], x[2], x[1], x[3]\} = \{1, 4, 3, 5\}$. Stage 1: $1 \pm 4 = 5, -3$ and $3 \pm 5 = 8, -2$, giving $\{5, -3, 8, -2\}$. Stage 2 with $W_4^0 = 1, W_4^1 = -j$: $5 \pm 8 = 13, -3$ and $-3 \pm (-j)(-2) = -3 \pm j2$, giving $\{13, -3+j2, -3, -3-j2\}$. ✓ $X[0] = 1 + 3 + 4 + 5 = 13$.
- **75 Ch asks for $X(5)$ of a 4-point DFT.** The DFT is periodic in k with period N , so $X(5) = X(5-4) = X(1) = -2+j4$, and $X(3) = -2-j4 = X^*(1)$. **That periodicity is the whole question**; computing a 5th value would be a mistake, not extra credit.
- **76 Bh's constant sequence transforms to an impulse**: a flat signal has energy at DC and nowhere else. Free check on a DIT you can do in your head.

1.2 Papers whose sequence is a formula, an impulse, or missing

HOT 6/45 78 Ch **7**, 77 Ch **8**, 75 Bh **2+8**, 71 Ch, 70 Asa **2+6**, 79 Bh **2+6**

- **Sample the formula first**, then run the ordinary recipe. All three are 8-point DIF.

Paper	Formula	$x[n], n = 0 \dots 7$	$X(k)$
78 Ch	$0.5 \sin(n\pi/6)$	$\{0, 0.25, 0.433, 0.5, 0.433, 0.25, 0, -0.25\}$	$\{1.616, -0.9633-j0.9633, -j0.25, 0.09732-j0.09732, 0.116, 0.09732+j0.09732, j0.25, -0.9633+j0.9633\}$
77 Ch	$\sin(3\pi n/8)$	$\{0, 0.9239, 0.7071, -0.3827, -1, -0.3827, 0.7071, 0.9239\}$	$\{1.497, 2.848, -2.414, -0.8478, -0.6682, -0.8478, -2.414, 2.848\}$
75 Bh	$0.2n$	$\{0, 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4\}$	$\{5.6, -0.8+j1.931, -0.8+j0.8, -0.8+j0.3314, -0.8, -0.8-j0.3314, -0.8-j0.8, -0.8-j1.931\}$

- **77 Ch's spectrum is entirely real**, because its sampled sequence happens to be symmetric about $n = 4$. A real spectrum is a legitimate answer; do not hunt for a missing j .
- **75 Bh asks only for $X(4)$ and $X(7)$.** $X(4) = -0.8$ is the alternating sum $0 - 0.2 + 0.4 - 0.6 + 0.8 - 1 + 1.2 - 1.4$; $X(7) = X^*(1) = -0.8 - j1.931$. **Two butterflies, not twenty-four.**
- **71 Ch and 70 Asa** give $x(n) = \{1, 0, 0, 0, 0, 0, 0, 0\}$, a unit impulse, and ask for $X(7)$. Every $X(k) = 1$, so **$X(7) = 1$** . The flow graph is still worth drawing, because that is where the marks are: the answer itself is one line.
- **79 Bh prints no sequence at all.** The paper says only "find the 8-point DFT of the following sequence using radix-2 DIT-FFT", and nothing follows. Verified on both available scans. **Answer the method**: draw the general 8-point DIT flow graph, state the bit-reversed input order and the three stages, and say the numeric answer needs the data that is missing.

1.3 72 Ash: the IDFT through a forward FFT

PIN 1/45 72 Ash 2+6

QUESTION AS PRINTED 2072 Ashwin · [2+6]

Use the FFT algorithm to compute the IDFT of the sequence $X(k) = \{6, -2 + 2j, -2, -2 - 2j\}$.

- Use $x[n] = \frac{1}{N} (\text{DFT}\{X^*[k]\})^*$ from §7.3.
- **Conjugate:** $X^*(k) = \{6, -2 - 2j, -2, -2 + 2j\}$.
- **Forward 4-point DFT** of that, by DIT: bit-reversed input $\{6, -2, -2 - 2j, -2 + 2j\}$; stage 1 gives $\{4, 8, -4, -4j\}$; stage 2 with $W_4^1 = -j$ gives $\{0, 4, 8, 12\}$.
- **Conjugate and divide by $N = 4$:**

$$x[n] = \frac{1}{4} \{0, 4, 8, 12\}^* = \{0, 1, 2, 3\}$$

- Check forwards: $\sum x[n] = 6 = X(0)$. ✓ And $x[n]$ came out real, as it must, because $X(k)$ was conjugate symmetric ($X(3) = X^*(1)$).

2 Circular convolution

TOP 39/45 15 papers direct, 12 as $x_3 = \text{IDFT}(X_1 X_2)$, 4 as a linear convolution · 253 marks, and all 31 problems are one calculation

QUESTION AS PRINTED 2080 Bhadra · 2080 Baishakh · [6] [2+5]

Compute the circular convolution of $x_1 = \{1, 2, 3, 1\}$ and $x_2 = \{4, 3, 2, 2\}$.

Route A: the matrix, which is what to do in the exam

- $N = 4$. Build the circulant of x_2 : row n , column m holds $x_2[((n - m))_4]$, so each row is the previous one shifted right by one.

$$\begin{bmatrix} 4 & 2 & 2 & 3 \\ 3 & 4 & 2 & 2 \\ 2 & 3 & 4 & 2 \\ 2 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+4+6+3 \\ 3+8+6+2 \\ 2+6+12+2 \\ 2+4+9+4 \end{bmatrix} = \begin{bmatrix} 17 \\ 19 \\ 22 \\ 19 \end{bmatrix}$$

$$y[n] = \{17, 19, 22, 19\}$$

- **Check:** $\sum y = 77$, and $(\sum x_1)(\sum x_2) = 7 \times 11 = 77$. ✓ **Two additions and a multiplication.**

Route B: through the DFT, if the paper insists

- $X_1(k) = \{7, -2 - j1, 1, -2 + j1\}$ and $X_2(k) = \{11, 2 - j1, 1, 2 + j1\}$.
- Multiply term by term: $X_3(k) = \{77, -5, 1, -5\}$.
- IDFT with W_4^{-nk} , i.e. $1, j, -1, -j$ as nk runs $0, 1, 2, 3$:

$$y[0] = \frac{1}{4} (77 - 5 + 1 - 5) = \frac{68}{4} = 17$$

$$y[1] = \frac{1}{4} (77 + (-5)(j) + (1)(-1) + (-5)(-j)) = \frac{76}{4} = 19$$

$$y[2] = \frac{1}{4} (77 + (-5)(-1) + (1)(1) + (-5)(-1)) = \frac{88}{4} = 22$$

$$y[3] = \frac{1}{4} (77 + (-5)(-j) + (1)(-1) + (-5)(j)) = \frac{76}{4} = 19$$

The imaginary parts cancel in pairs at every n , as they must for a real answer. Same $\{17, 19, 22, 19\}$.

- **This route costs three transforms to reach a result the matrix gives in one table.** Use it only when the question names the DFT.

Every other direct circular convolution

Papers	N	x_1	x_2	$y[n]$
81 Ba	4	{1, 2, 4, 5}	{2, 1, 6, 8}	{47, 67, 56, 34}
72 Ch, 82 Bh	5	{1, 2, 3, 1}	{1, 2, 1, -1, 1}	{1, 6, 9, 8, 4}
79 Ba	5	{1, -1, -2, 3, -1}	{1, 2, 3}	{8, -2, -1, -4, -1}
82 Ba	5	{1, 1, -1, -1, 2}	{1, -1, -2}	{1, -4, -4, -2, 5}
78 Bh, 74 Ma	4	{2, 1, 2, 1}	{1, 2, 3, 4}	{14, 16, 14, 16}
78 Ch	4	{2, 1, 2, 1}	{1, -2, -1, 3}	{1, 2, 1, 2}
75 Ash	4	{1, 2, -1, 1}	{1, 3, 5, 7}	{13, 3, 17, 15}
75 Ch	4	{0, 0, 1, 1}	{1, 1, 1, 1}	{2, 2, 2, 2}
76 Bh	4	{1, 0.5, 1, 0.5}	{0, 1, 2, 3}	{4, 5, 4, 5}
71 Bh	5	{1, 2, 0, 3, 4}	{2, -1, 2, -1, 2}	{8, 8, 2, 17, 5}
67 Mng	4	{1, 2, 1}	{1, 2, 0, 1}	{3, 5, 5, 3}
66 Ma	4	{1, 2}	{1, 3, 4, 5}	{11, 5, 10, 13}
69 Bh	4	{1, 0, 1}	{1, 0, 2, 1}	{3, 1, 3, 1}
71 Ch, 70 Asa, 79 Ch	4	{1, 2}	{1, 1, 1, 1}	{3, 3, 3, 3}

75 Ch's answer is constant because x_2 is constant: convolving anything with a flat sequence circularly gives $\sum x_1$ in every position. Same reason 71 Ch's answer is {3, 3, 3, 3} with $\sum x_1 = 3$. Both are free marks once you see it.

2.1 "Find $x_3[n]$ if $X_3(k) = X_1(k)X_2(k)$ "

TOP 12/45 76 Ch, 69 Ch, 72 Ma, 79 Bh, 81 Bh, 81 Ch, 75 Bh, 73 Ch, 73 Shr, 80 Ch, 73 Bh, 73 Ma, 74 Ch, 72 Ka, 71 Shr · **this IS the circular convolution; take no transform at all**

By the convolution property of §7.2, multiplying the DFTs is circular convolution in time. Write "by the circular convolution property, $x_3[n] = x_1[n] \otimes x_2[n]$ ", then build the matrix. **One line of theory replaces three DFTs and an IDFT.**

Papers	N	x_1	x_2	$x_3[n]$
76 Ch, 69 Ch	4	{1, 2, -2}	{1, 2, 3, -1}	{-7, 6, 5, 1}
72 Ma	4	{1, 2, 3, -1}	{2, 1, -3}	{-8, 8, 5, -5}
79 Bh	4	{1, 0, 0, 1}	{2, 0, 2}	{2, 2, 2, 2}
72 Ka	4	{1, 2, 4}	{-1, 2, 3, 1}	{13, 4, 3, 15}
71 Shr	4	{1, 2, 3, 4}	{1, 3, 5, 7}	{42, 46, 42, 30}
81 Bh, 81 Ch, 75 Bh	5	{1, 3, 9, 27}	{1, 2, 4, 8, 16}	{229, 365, 451, 65, 130}
73 Ch	5	{1, -2, 5, 1, 2}	{1, 2, -3, -2}	{-8, -8, -6, 15, -7}
73 Shr	5	{1, -2, 2, 1, 4}	{2, 1, -3, -1}	{1, -16, -5, 9, 5}
80 Ch, 73 Bh	5	{1, 2, 3, -1, 5}	{2, 1, -3}	{10, -10, 5, -5, 0}
73 Ma	5	{3, 2, 1, -1, -2}	{1, 2, 3}	{-4, 2, 14, 7, -1}
74 Ch	5	{1, 2, 0, 1, -2}	{1, 0, 1, 1, 2}	{6, 1, 1, 0, 2}

- **71 Shr is the same question in disguise again:** "input $x[n]$, impulse response $h[n]$, find $y[n]$ if $G[k] = X[k]H[k]$ ". It is the 4-point circular convolution, {42, 46, 42, 30}, and **not** the linear one, because no padding was asked for.
- **79 Bh's answer is constant** for the same reason as §2: x_1 padded is {1, 0, 0, 1} and x_2 is {2, 0, 2, 0}; every output picks up $\sum x_1$ · the same pairing.

The defect: 81 Ch asks for a 4-point DFT of a 5-point sequence **1+6 1+5**

81 Ch, 75 Bh and 81 Bh set identical sequences and only one states a workable N

- All three give $x_1[n] = (3)^n$ for $0 \leq n \leq 3$ (four values, {1, 3, 9, 27}) and $x_2[n] = (2)^n$ for $0 \leq n \leq 4$ (**five values**, {1, 2, 4, 8, 16}).
- **81 Bh** says "5-point DFT". Fine, and the answer is the table row above.
- **81 Ch** says "**4-point** DFT". A 4-point DFT of a five-entry sequence **does not exist**; there is nowhere for $x_2[4] = 16$ to go. As printed the question is unanswerable.
- **75 Bh** states **no N at all**. With none given, N must be at least $\max(L, M) = 5$, so it is 81 Bh's question.
- **What to write:** take $N = 5$, say in one line why (a 5-entry sequence needs $N \geq 5$; the paper's own "what is zero padding" part is the justification), and give $x_3 = \{229, 365, 451, 65, 130\}$. **Naming the defect is worth more than forcing an answer.**
- Note for the PYQ document: its heading groups 75 Bh with 81 Ch under "4-pt". That is an inference from 81 Ch's wording, and it is the wrong one.

2.2 Linear convolution through circular convolution

HOT 4/45 74 Ch **1+5**, 74 Bh **2+6**, 72 Ash **5**, 77 Ch **6+2** · the only difference is the choice of N

QUESTION AS PRINTED 2074 Chaitra · [1+5]

What is zero padding? Find the linear convolution of $x = \{1, 1, 1, 1\}$ and $h = \{2, 3\}$ using circular convolution.

- $L = 4$, $M = 2$, so the linear result has $L + M - 1 = 5$ samples. **Zero pad both to length 5**: $x = \{1, 1, 1, 1, 0\}$, $h = \{2, 3, 0, 0, 0\}$.
- The 5×5 circulant of h against x gives

$$y[n] = \{2, 5, 5, 5, 3\}$$

- Check: $\sum y = 20 = (\sum x)(\sum h) = 4 \times 5$. ✓
- **Why the padding matters, in one demonstration.** Convolving the same two sequences circularly at $N = 4$ instead gives $\{5, 5, 5, 5\}$. The 5th sample, 3, had nowhere to go, so it **wrapped onto** $y[0]$: $2 + 3 = 5$. That is the time-domain aliasing the whole question is about, and it is the answer to the "what is zero padding" part.

Paper	x	h	N	$y[n]$
74 Ch	$\{1, 1, 1, 1\}$	$\{2, 3\}$	5	$\{2, 5, 5, 5, 3\}$
74 Bh	$\{-1, 1\}$	$\{2, 3, 1, -2\}$	5	$\{-2, -1, 2, 3, -2\}$
72 Ash	$\{1, 1, 1\}$	$\{2, 2, 2\}$	5	$\{2, 4, 6, 4, 2\}$
77 Ch	$\{1, 1, 1\}$	$\{1, 0, -3\}$	5	$\{1, 1, -2, -3, -3\}$

- **77 Ch's $x[n]$ is misprinted** as " $u[n] - u[n-3]$ ". It means $u[n] - u[n-3] = \{1, 1, 1\}$, and $h[n] = \delta[n] - 3\delta[n-2] = \{1, 0, -3\}$.
- **72 Ash's answer is a triangle**, $\{2, 4, 6, 4, 2\}$, because convolving a rectangle with a rectangle always gives a trapezoid, and with two equal lengths it degenerates to a triangle. A shape check worth two seconds.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads "leads to / therefore", $+$ reads "and", $=$ reads "is". ✓ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.